Physical Account

Every number we rely on beyond Annex A and B, every Annex value we changed, and all we invented. Reasons and uses: detail page.

Annex C row	What we assumed	Value	Why	Used in
Electrical power	guest reserve	cruise 200, one set down 100 kW; slow 250, fast 150; admit at 200	propulsion out of scope	code, test
	peak tolerance	peaks never credited; gate min(100, 120) - 5 headroom = 95 kW	peaks last 60 s (DEC-35)	code, test
	mission lines	port: Bay 1, mast; stbd: Bay 2, deck, roof; line lost on a trip: none	total reserve binds	code, test, documented only
	plant check	gensets at 0.9; hotel 400 kW; aux 55 kW; trip sheds 150 kW of hotel	consistency check only	test
Heat rejection	loop, seawater	-40 % = 1 of 2 pumps; chillers essential; scan 0.1 °C steps to 35 °C	cooling survives a trip	test, documented only
Positions	hull, roof, ramp	60 m hull; roof holds G5's antenna (60 kg); ramp craft up to 7 m	roof clears drones	card, code
Mass / stability	G4 hangar	850 kg, not Annex B's 900 kg	organizer fix (DEC-22)	card, test
	Scenario B set	G3, G4, G5, 9.515 t; brief's G2-G5 (13.015 t of 10) refused	override logged (DEC-32)	test
Data links	rates judged	guests' minimum rates, nominal gap shown as degraded; SATCOM at its 2 Mbit/s floor; floors G3 acoustic 1 kbit/s, G5 SATCOM 1 Mbit/s	admission rule (DEC-38)	card, code, test, documented only
	LOS radio	top rate 20 Mbit/s, G4 kept within 20 km (A.5 reach 20-40 km)	unverified; range is ours	code, test
	acoustic link	up only while course steady, noise quiet, no deck machinery; lost at once; 60 s per message; Scenario B water not layered	A.5's rule, safe end	code, test, documented only
State transitions	R3 lead time	confirm 1 min at 14, 3 at 6; clear deck 5 at 14, 12 at 6	A.6 slow end, worst case	code, test
	Scenario B seed	confirm 1, clear 4, approach 4 min: G4 lands at minute 17	replayable, inside A.6	test
Compute / response time	fault entry	trip on 2 of 3 signals, pump on alarm plus flow or delta-T, held 1 s; restore after 60 s; a feed silent 3 s falls to its conservative value	one reading never sheds	code, test
	timeout, safety	on a miss: admit no, hold, run the ready ladder; safety by table lookup	no solver on 200 ms	code, documented only
	compute	cores G1 2, G2 2, G3 4, G4 4; 4 boards, 6 Pi nodes	Annex B gives no compute	card, code, documented only, planned
Mission priorities / crew	R3 deck rules	14: officer reports, bridge confirms; 6: standing plan from launch; 0: only on a deck latched clear at the last sweep, else forbidden	no one clears deck at 0	code, test, documented only
	R3 timing	approach 4 min; drain 1.11 %/min; stale 10 s; secured 60 s	drain unverified	code, test
	mission weights	high 1.0, medium 0.5, low 0.2, ours if unstated; 1-2 critical each	Annex B: high/medium/low	code, missions, test
Anything else	guest modes	kW/kept: G2 sampling 6/0.8, standby 1/0; G3 standby 1/0.3; G4 charge 10/0.7, standby 2/0.3; G3 cradle 3.5 t, vehicle 2 t 5.5 m	Annex B gives only kW	card
	invented	G6 card; pump fault at minute 19: loop 80 kW, ceiling 75, load 81	cooling binds (DEC-40)	card, code, test

Closing question. One change: a standby chilled-water pump, so losing a pump no longer costs 40 % of the loop (A.2), 48 kW at 30 °C. Our minute-19 pump fault then sheds nothing: at 24 °C the loop keeps 133.3 kW, not 80, so the 81 kW load fits under 95 kW, not 75. At 30 °C a pump loss today leaves guests 67 kW for up to 14 days with no repair at sea (A.6, A.10), against 105 kW for our Scenario B set and 100 for A5; with the pump both fit under 115 kW. An exchanger fault still costs 40 %. A 50 kW battery for a genset restart helps less: at 30 °C cooling caps it at 20 kW.